Learning to Recognise Gravitationally Lensed Fast Radio Bursts Without Real Positives

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***Abstract*—A fast radio burst (FRB) lensed by a foreground galaxy reaches Earth twice, days to months apart, and the two copies differ only by propagation effects. Recognising such copies among the bursts of repeating sources is a pairwise verification problem with three difficulties: no real positive example exists, the negatives include adversarial *impostors* that share a propagation imprint, and decisions must hold at a 1% false-positive rate. We build a semi-synthetic benchmark from all 981 public dynamic spectra of the 83 repeating sources in the CHIME/FRB Catalog 2, and propose a 259k-parameter Siamese network that aligns the two bursts' feature maps with a differentiable correlation layer before fusing them. In a single pre-registered blind test over all 80 eligible sources, each scored by a model that never saw it, the network recovers 0.436 [0.226, 0.675] of lensed pairs at 1% false-positive rate against 0.211 [0.153, 0.293] for the best physics- or feature-based baseline, a paired gain of +0.226 [+0.067, +0.403]; on the 20 sealed test sources alone the gain is +0.323 [+0.115, +0.543]. Ablations show that both alignment and impostor training are needed. Because the positive class has no real counterpart, we also validate on real data: 18585 real cross-source pairs, which are true negatives, give a false-positive rate of 1.0% [0.5, 1.8] against per-pair simulated twins, two copies of a real burst are recognised as often as two copies of its template, and the pipeline reproduces on a second machine. Finally, survey geometry, not the classifier, limits discovery: a transit telescope records both copies for only about 2% of lenses, and an all-sky monitor would find about 50 times more lensed repeaters from the same sources.**

***Index Terms*—Siamese networks, pairwise verification, simulation-to-real transfer, deep learning for signal processing, radio astronomy**

I. Introduction

Fast radio bursts are millisecond radio flashes from distant galaxies [1], [2], and some sources repeat thousands of times. When a galaxy lies close to the line of sight, gravitational lensing forms two images of the source, so every burst arrives twice, separated by a fixed delay Δt of days to months (Fig. 1a). A confirmed lensed FRB would measure the expansion rate of the Universe and test dark-matter models [3]–[5], but none has been found [6]–[9].

Finding one is a pairwise verification problem: are two bursts from the same sky position two copies of one burst, or two different bursts of a repeating source? Three properties make it hard. *No real positives:* any detector is trained on simulation, and its transfer to real data must be checked without labelled positives. *Adversarial negatives:* distinct bursts of one source can share a spectral envelope imposed by a plasma lens, or a scintillation pattern, and resemble copies. *Low operating point:* the probability that a given repeater is strongly lensed is 10^{-4} or less (the lensing optical depth is 1.5×10^{-4} even at redshift 1), so decisions must hold at 1% false-positive rate or below.

This paper makes four contributions (Fig. 2):

1) a semi-synthetic pair benchmark built from all public CHIME/FRB Catalog 2 repeater spectra, with impostor negatives and frozen, hash-checked evaluation sets;
2) a Siamese correlation network that aligns feature maps before comparing them, with an ablation study over three seeds;
3) a pre-registered evaluation for a detector without real positives: one blind test over all sources, true-negative transfer with per-pair twins, a real-burst positive proxy and an independent reproduction;
4) an analysis of where such a detector can work, showing that transit telescopes rarely record both copies and that sky coverage, not classifier accuracy, limits discovery.

II. Related Work

Siamese verification. Siamese networks decide whether two inputs share an identity by comparing the outputs of one shared encoder, from signature verification [10] and contrastive embedding [11] to one-shot recognition [12]. Most compare global embedding vectors, which discards where features occur; patch-matching studies contrast such networks with two-channel networks that fuse the inputs early [13], and our baselines include both. Comparing feature maps over a range of displacements was introduced for optical flow [14] and used for Siamese tracking [15]; we use it to estimate and remove the residual time offset between two bursts before comparing them.

Lensed-pair identification. For gravitational waves, lensed event pairs have been identified with a DenseNet and gradient boosting [16], and with a Siamese residual network that fuses features by absolute difference and product [17], which notes the need to adapt to real detector data. FRB lensing searches have used auto- and cross-correlation statistics [6], [7], [9], and machine learning on FRBs has separated repeating from non-repeating sources [18]; to our knowledge no learned detector of lensed FRB copies exists. Detectors trained on simulation

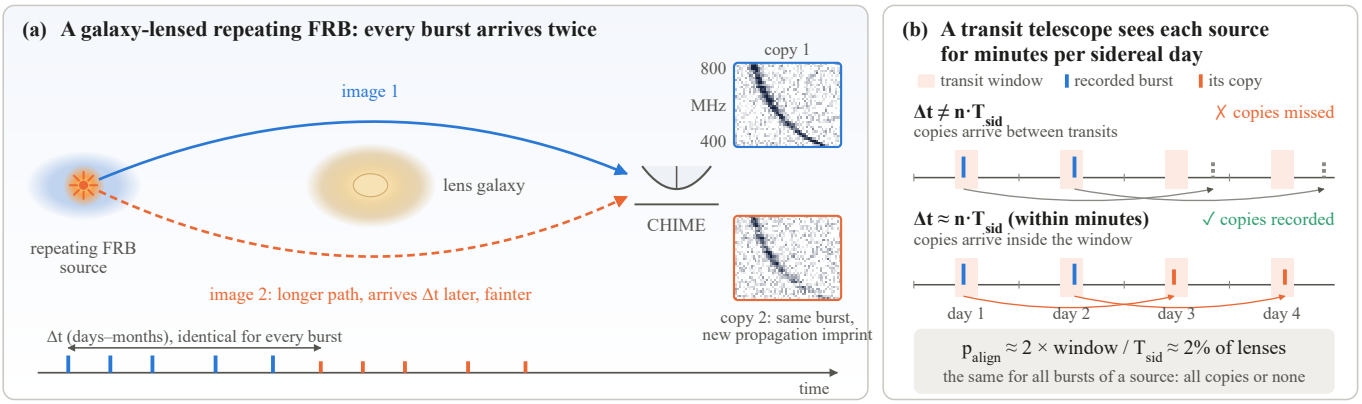


Fig. 1. The problem. (a) A galaxy lens sends every burst of a repeating FRB along two paths; the copy arrives a fixed Δt later, fainter, with its own scintillation and scattering. The dynamic spectra show one burst sweeping from 800 to 400 MHz (the ν^{-2} dispersion delay), computed from the cold-plasma law. (b) A transit telescope such as CHIME records a source only within minutes of transit, so the copies of recorded bursts are recorded only if Δt is close to a whole number of sidereal days.

are made robust by randomising nuisances [19], but transfer still has to be shown on real data.

III. Problem Formulation

A burst is a dynamic spectrum $X \in \mathbb{R}^{F\times T}$ with $F = 256$ frequency channels and $T = 160$ time samples of 0.983 ms, normalised to unit off-pulse noise. A pair (A, B) of bursts from one sky position has a label $y \in \{\mathrm{L}, \mathrm{R}, \mathrm{U}, \mathrm{I}\}$: lensed copies, repeat (two bursts of one source), unrelated (two sources) or impostor (two bursts of one source sharing a propagation imprint). A detector outputs a symmetric score $s(A, B) = s(B, A)$, and a pair is flagged if $s > \tau_\alpha$, where τ_α is the $(1-\alpha)$ quantile of s over negatives. The primary metric is

$$\mathrm{Recall@}\alpha = \Pr[s(A, B) > \tau_\alpha \mid y = \mathrm{L}], \quad \alpha = 1\%, \qquad (1)$$

with negatives $\{\mathrm{R}, \mathrm{I}\}$, the alternatives that a real search faces within one source. Pairs share sources, so we report 95% intervals from a bootstrap over sources that recomputes τ_α in every resample. The validation constraint is that no real pair with $y = \mathrm{L}$ is available: real data can only supply negatives.

IV. Benchmark Construction

Real inputs. We use the Stokes-I dynamic spectra of all 981 bursts of the 83 repeating sources in CHIME/FRB Catalog 2 [20], recorded by the CHIME transit telescope [21] at 400–800 MHz. Each burst is binned to 256×160 around its emission and normalised by its off-pulse noise. The catalogue's fitted burst models serve as noise-free templates (950 usable), and the off-pulse columns of each file, mirror-tiled, serve as real noise backgrounds (856), each with its own channel mask.

Pair classes. A lensed pair places one template twice; a repeat pair, two templates of one source; an unrelated pair, templates of two sources. An impostor pair is a repeat pair in which member B inherits member A's 25-MHz-smoothed spectral envelope (I1, mimicking a plasma lens), both members share one scintillation pattern (I2), or both. Each member is injected at a measured matched-filter signal-to-noise ratio (S/N) on a different real background, with its own time shift (± 10 samples), residual dispersion, scattering and scintillation, so that only burst identity separates lensed from repeat pairs. Probes on single-member statistics cannot tell the classes apart (AUC ≤ 0.53).

Splits and frozen sets. Sources are split 60/20/20 into train, validation and test. Validation and test sets of 4000 pairs, and "bright" sets with pair minimum S/N ≥ 25 (the pre-registered headline bin), are regenerated from fixed seeds and checked against stored SHA-256 hashes. The test split stays sealed under a pre-registered blind protocol; all numbers below are development results on sources never used in training.

V. Siamese Correlation Network

Encoder. Inputs pass through asinh to compress the dynamic range of bright bursts. A shared encoder g of four 5×3 convolution blocks with group normalisation maps X to $g(X) \in \mathbb{R}^{C\times F'\times T'}$ with $C = 64$, $F' = 16$ and $T' = 80$, keeping two-sample time resolution because lensed copies differ by residual time offsets.

Correlation and soft alignment. For lags $\ell \in \{-L, \ldots, L\}$ with $L = 12$ encoder steps (± 24 samples),

$$\rho(\ell) = \frac{\langle \bar{g}_A,\ \bar{g}_B(\ell)\rangle}{\|\bar{g}_A\|\,\|\bar{g}_B(\ell)\|}, \qquad w_\ell = \frac{e^{\beta\rho(\ell)}}{\sum_k e^{\beta\rho(k)}}, \qquad (2)$$

where $g_B(\ell)$ is g_B shifted by ℓ in time, the bar denotes removal of the time mean, and β is learned. The aligned map is $\tilde{g}_B = \sum_\ell w_\ell\, g_B(\ell)$; this is a differentiable estimate of the residual delay that the network can sharpen or soften.

Fusion and head. The fused map $[\,|g_A - \tilde{g}_B|,\ g_A \odot \tilde{g}_B\,]$ passes two convolution blocks and mean and max pooling; an MLP receives the pooled features together with ρ and outputs four logits z. The lensed score is the log-odds against the

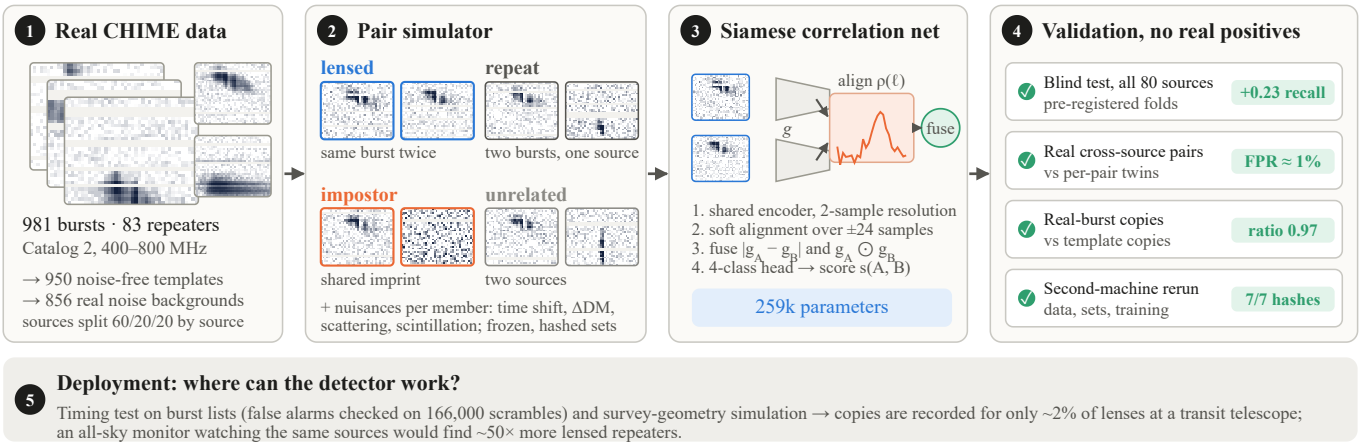


Fig. 2. Workflow. (1) Real CHIME/FRB Catalog 2 dynamic spectra (tiles: real bright bursts) provide noise-free templates and real noise backgrounds. (2) The pair simulator places templates on different real backgrounds with independent nuisances to make four classes; impostors share a spectral envelope or scintillation pattern. (3) The Siamese correlation network scores each pair. (4) Validation uses only real negatives, a real-burst proxy and an independent rerun. (5) Survey-geometry analysis decides where the detector can be deployed.

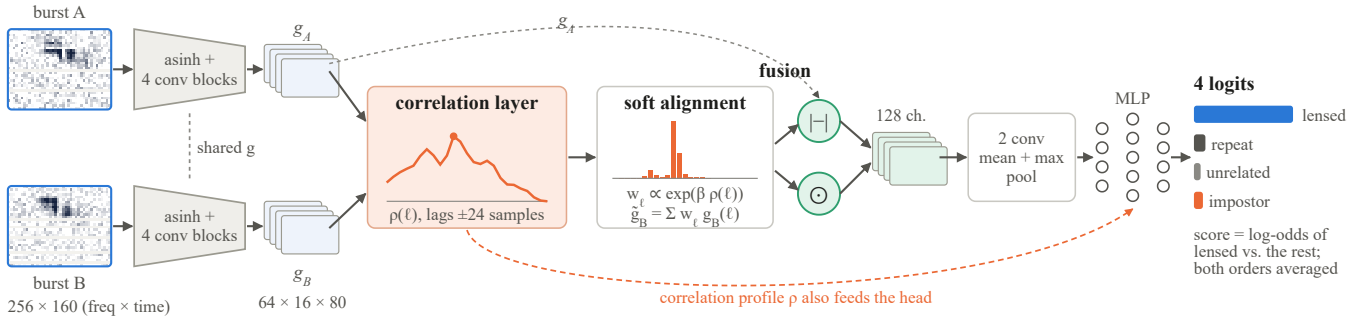


Fig. 3. Model 6. A weight-shared encoder maps each burst to a $64 \times 16 \times 80$ feature map at two-sample time resolution. The correlation layer measures the normalised correlation $\rho(\ell)$ over lags of ± 24 samples (curve: computed for the pair shown), softmax weights align g_B to g_A, and the aligned maps are fused by absolute difference and product. The correlation profile also feeds the head. Both directions are computed and averaged.

other classes, and symmetry is enforced by averaging both directions:

$$s(A,B) = \tfrac{1}{2}\left[\sigma(A,B) + \sigma(B,A)\right], \quad \sigma = z_{\mathrm{L}} - \log \sum_{c \neq \mathrm{L}} e^{z_c}. \tag{3}$$

The model has 259k parameters.

Training. All learned models are trained for 20 000 steps of 64 pairs drawn on the fly from training sources (half with S/N floor 10, half 25), with four-class cross-entropy, AdamW (learning rate 10^{-3}, weight decay 10^{-4}), 1000 warm-up steps and cosine decay; the checkpoint is selected on the validation bright set.

VI. Evaluation Without Real Positives

Pre-registered blind test. Every choice below was fixed in a written plan before the result it affects. The primary comparison is model 6 against the best of methods 1–3 on the validation set, which is the GBM, and model 6 wins only if the lower 95% bound of the paired recall difference is above zero. Because the validation interval was wide (half-width 0.16), a pre-registered trigger replaced the fixed test split by five source-grouped folds over all 80 sources with at least two templates (16 per fold), with frozen settings, one seed per fold and the final checkpoint, so that every source is scored by a model that never saw it; a rehearsal on the 60 development sources gave +0.22 [+0.04, +0.38]. The bootstrap uses a pooled threshold across folds, which penalises score-scale differences between fold models. The test sources were opened once, after all methods were frozen at a version tag. One deviation is disclosed: the test was opened before the nuisance ranges of the propagation model were formally signed off, so a later change of those ranges would require a new, separately reported test. The fixed-split test sets (20 sources) serve as secondary checks.

True-negative transfer. Two bursts from different sources cannot be lensed copies, so real cross-source pairs are labelled negatives that can be scored without breaking the blind protocol. For each real pair we simulate five *twins* from the same two templates at their measured S/N and set τ at the twins' 99th percentile; the pre-registered criterion is a real false-positive rate of at most 2% per S/N bin. Twin matching matters: with generic simulated pairs as reference, cross-

TABLE I

RECALL AT 1% FPR (LENSED VS REPEAT + IMPOSTOR), S/N $\geq$ 25, 95% SOURCE BOOTSTRAP. VALIDATION: DEVELOPMENT, CHECKPOINTS SELECTED ON IT. TEST SOURCES: SEALED UNTIL THE SINGLE TEST RUN. ALL 80 SOURCES: THE PRE-REGISTERED PRIMARY, OUT-OF-FOLD (OOF) OVER FIVE SOURCE-GROUPED FOLDS, FINAL CHECKPOINTS.

Method	Validation (26 sources)	Test sources (20 sources)	All 80 sources (grouped-fold OOF)
(1) Cross-correlation	0.171	0.128	–
(2) Physics fit	0.164	0.070	–
(3) Feature GBM	0.285 [0.180, 0.401]	0.226 [0.152, 0.390]	0.211 [0.153, 0.293]
(4) Two-channel CNN	0.423 / 0.409 / 0.407	–	[OofFour]
(5) Classic Siamese	0.500 / 0.559 / 0.541	–	0.368 [0.190, 0.566]
(7) Cross-attention	0.471 / 0.013[a] / 0.579	–	[OofSeven]
(6) Ours	**0.701 / 0.698 / 0.684**	**0.596 / 0.459 / 0.594**[b]	**0.436 [0.226, 0.675]**
(6) − (3), paired	+0.394 [+0.226, +0.542]	+0.323 [+0.115, +0.543]	**+0.226 [+0.067, +0.403]**
AUC, (6) / (3)	0.987 / 0.925	0.976–0.985 / 0.934	0.982 / 0.936

[a]One of three seeds collapsed to a plateau. [b]Model 6 seeds retrained on a second machine, whose validation recall is 0.05 lower (0.668 / 0.649 / 0.616), so this column is conservative for model 6.

correlation seems to fail below S/N 25 (6.2, 3.8, 2.4%), yet its twins fail as often (5.6, 4.0, 2.5%), because those real pairs are brighter than generic ones; the failure is sample composition.

Real-burst positives. As a proxy for lensed pairs, two copies of a real burst's cutout, with its fine structure and real noise, are injected with independent nuisances on different backgrounds, alongside template copies and noise-only copies drawn with the same random numbers.

Reproduction. A second machine re-downloaded and re-processed all 981 spectra, rebuilt the injection cache, regenerated the frozen sets, retrained model 6 with three seeds and reran the checks.

VII. EXPERIMENTS

Baselines. All methods score the same pairs, averaged over both member orders: (1) normalised cross-correlation in the on-pulse box, maximised over ± 24 lags, with 16 channel blocks; (2) a physics statistic that fits one member to the other over scattering time, residual dispersion, lag and amplitude; (3) gradient-boosted trees (GBM) on seven measured burst properties per member, the best of 1–3 and the pre-registered comparator; (4) a two-channel CNN; (5) a classic Siamese network comparing embedding vectors; and (7) cross-attention between time–frequency patches.

Main result. The pre-registered criterion is met: over all 80 sources model 6 recovers 0.436 [0.226, 0.675] of lensed pairs against 0.211 [0.153, 0.293] for the GBM, a paired gain of +0.226 [+0.067, +0.403] whose lower bound is above zero (Table I). With a separate threshold per fold the recalls are 0.506 and 0.226, so the pooled threshold costs model 6 more than the GBM. The gain agrees with the development rehearsal (+0.22 [+0.04, +0.38]) and appears on every set: +0.39 on validation and +0.323 [+0.115, +0.543] on the 20 sealed test sources, where model 6 is represented by second-machine seeds. Every learned model beats the physics and feature baselines. A gap of this size at similar AUC (0.982 against 0.936) shows that the gain lies in the low-false-positive tail that a search uses. With 16 sources per fold, recall varies between

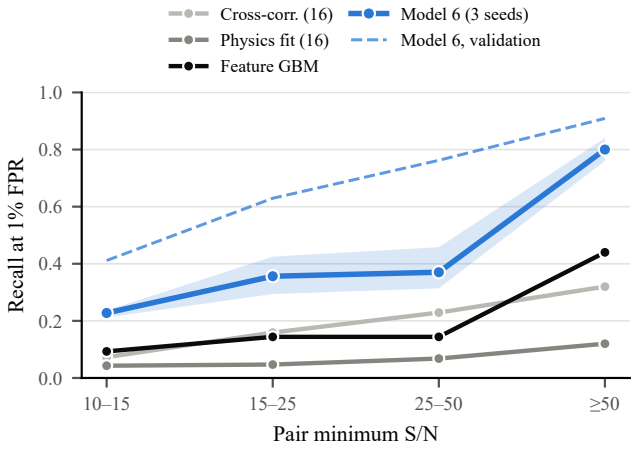


Fig. 4. Recall at 1% FPR by pair minimum S/N on the sealed test set (20 test sources; 537, 320, 118 and 25 lensed pairs per bin), from the single test run. Model 6: mean and range of three seeds; dashed: the same seeds on the validation set.

TABLE II

ABLATIONS OF MODEL 6 ON THE SECOND MACHINE, THREE SEEDS EACH: SEED-MEAN RECALL AT 1% FPR AND PAIRED DIFFERENCE FROM THE FULL MODEL (VALIDATION BRIGHT SET, 95% SOURCE BOOTSTRAP).

Variant	Recall	Difference
Full model	0.644	–
No alignment (correlation off)	0.557	−0.087 [−0.186, −0.025]
No impostors in training	0.474	−0.170 [−0.254, −0.052]
Hard-negative repeats	0.698	+0.053 [−0.011, +0.082]

folds (0.19–0.86 for model 6, 0.16–0.29 for the GBM), which the source bootstrap carries into the interval.

Sensitivity and errors. On the test sources the gain holds in every S/N bin (Fig. 4). Model 6 and the GBM both recall less than on validation, as expected for sources that played no part in development (model 6's checkpoints and the choice of the GBM as comparator were both made on validation), while the untrained baselines move up or down. At the pooled 1% threshold of the primary test, 191 of 200 of model 6's false positives are impostors with a shared envelope (I1 or I1+I2) and 7 are plain repeats: the remaining confusion is the physically hardest case by design. Model 6 classifies 95% of lensed pairs of unseen sources correctly, while its confusion of repeat and unrelated pairs on unseen sources (32% correct against 86% on seen ones) shows that part of its capacity memorises source identity. Three training variants aimed at this memorisation (merging the repeat and unrelated targets, a random bandpass per member, or both) changed recall by -0.02 to $+0.00$, with every interval including zero, so the lensed decision does not depend on source identity.

Ablations. Removing the alignment, or training without impostors, lowers recall in every seed (Table II); without impostors the network mistakes envelope-sharing bursts of one source for copies. Oversampling template-similar repeats helps slightly in every seed, but the interval includes zero, and on

TABLE III
REAL-DATA VALIDATION OF MODEL 6: VALIDATION SOURCES ON TWO MACHINES, AND ALL 80 SOURCES WITH THE OUT-OF-FOLD MODELS OF THE TEST RUN (95% SOURCE BOOTSTRAP WHERE GIVEN).

Check (criterion)	Validation sources		All 80 sources
	Original	Second machine	(OOF models)
Real FPR at twin threshold, S/N $\geq$ 25 ($\leq$ 2%)	1.1 / 1.4 / 0.8%	1.5 / 1.3 / 1.0%	1.0% [0.5, 1.8]
GBM, same check	0.9%	1.0%	0.9%
Real–twin rank corr. / twin ceiling	0.91–0.94 / 0.88–0.91	0.91–0.92 / 0.87–0.89	0.91 / 0.90
Real-burst / template recall, S/N 25 ($\geq$ 0.8)	0.96 [0.89, 1.06]	0.97 [0.91, 1.08]	–
Real-burst / template recall, S/N 40	0.99 [0.91, 1.13]	1.04 [0.97, 1.18]	–
Noise-only copies flagged ($\leq$ 0.05)	0.000	0.000	–
Frozen-set hashes	reference	7/7 match	–

a closed-form benchmark this scheme biased a learned likelihood ratio low by about 1 dex, so we do not adopt it without a correction. Among the architectures on the 80-source folds, model 6 leads the classic Siamese network by +0.068 [−0.013, +0.154], the two-channel CNN by [OofSixMinusFour] and cross-attention by [OofSixMinusSeven]; every learned model still beats the GBM.

Real-data validation. On the 3851 real cross-source pairs of the validation sources, all 12 trained models and 3 baselines meet the twin-threshold criterion once short real windows are filled with off-pulse noise like the simulated backgrounds, and model 6's real-to-twin rank correlation reaches the twin-to-twin ceiling on both machines (Table III). Before the fill, model 6 flagged real pairs whose windows shared the same zero-padded edges; the twin test exposed this format mismatch, which the fill removes. Two copies of a real burst, with its fine structure, are recognised as often as two copies of its smooth template, and noise-only copies never are, so model 6 does not rely on template smoothness or shared noise; the classic Siamese network does (ratio 0.72–0.84). In the test run the same transfer check, with each fold's model scoring real pairs of its own held-out sources, passes for every method on all 80 sources (18585 pairs, at most 4000 per fold drawn at random to bound compute). Thresholds frozen on validation, by contrast, give a realised false-positive rate of 1.5–3.4% for model 6 on the test sources (GBM 1.5%), so a search must set its thresholds on real negatives.

Efficiency. Model 6 scores 1149 pairs per second in both orders on one RTX 5070 Ti (bfloat16), so all 78 701 same-source pairs of Catalog 2 take about 70 s.

VIII. Survey Geometry: Where Can the Detector Work?

A lensed repeater can also be found without a classifier. A galaxy lens fixes Δt to far better than the 4 ms catalogue timing, so two burst pairs (i, j), (k, l) of one source with $|\Delta t_{ij} - \Delta t_{kl}|$ below four combined timing errors already form a significant pattern. We calibrated this timing test on burst lists scrambled by whole sidereal days, which keeps each burst's hour angle, and confirmed its false-alarm rate on independent scrambles (1 of 166,000 held-out scrambles).

Both the classifier and the timing test need the copies to be recorded. CHIME is a transit telescope: every recorded burst arrives within a window of width $W(\delta) = 2\theta_{\rm EW}/(15^\circ\,{\rm h}^{-1}\cos\delta)$ around transit, with $\theta_{\rm EW} \approx 1.1^\circ$ measured from the real bursts. The copy of each burst arrives at the same hour-angle offset, $24\,{\rm h} \times {\rm frac}(\Delta t/T_{\rm sid})$, so copies are recorded for all bursts or none (Fig. 1b), and for a broad delay distribution

$$p_{\rm align} \approx 2W(\delta)/T_{\rm sid}, \tag{4}$$

doubled above $\delta = 70^\circ$, where lower transits add a window. Simulating singular-isothermal-sphere lenses with each source's redshift, burst times and exposure gives a median $p_{\rm align}$ of 1.8%, following (4) (Fig. 5). Repeating FRBs therefore do not escape the transit problem known for one-off bursts [22].

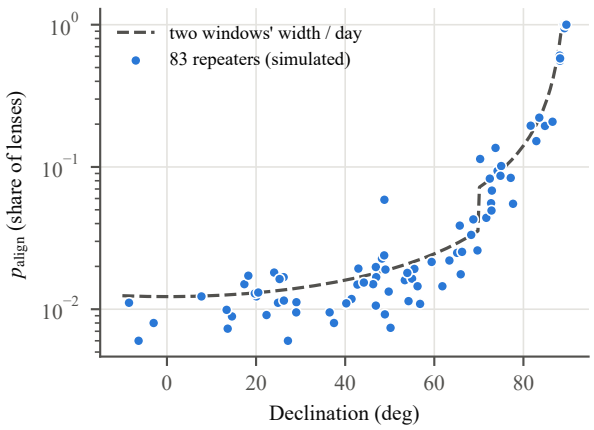


Fig. 5. Share of lens delays for which CHIME records the copies of a repeater's bursts: the 83 Catalog 2 repeaters (simulated with their own burst times, exposure and redshift) and (4) (dashed).

The expected number of detectable lensed repeaters in the Catalog 2 sample is 5×10^{-6} without and 1×10^{-4} with magnification bias (slope 1.5), quoted to one significant figure because independent reruns of the Monte Carlo differ by tens of percent; a null search is the expected outcome. A 10-year transit survey gains less than a factor of two, because the alignment requirement remains, while an all-sky monitor watching the same sources gains about 50. Raising the median host-galaxy dispersion measure from 68 to 150 pc cm^{-3} changes the yield by less than this scatter. For transit surveys, including next-generation drift-scan arrays such as CHORD [23], the classifier's role is to vet the rare single recorded copy and reject impostors; its full value lies with wide-field monitors such as BURSTT [24].

IX. Discussion and Limitations

Simulated false-positive rates rank methods, but deployment thresholds must come from real negatives, and twin matching is what makes such a check interpretable: it separates simulation gaps from differences in sample composition, and it exposed a real-versus-simulated format mismatch that no simulated benchmark would show. Pre-registration made the single test run decisive: the criterion, comparator, metric,

bootstrap and fold rules were fixed before the test sources were seen, and the gain on them matches the development estimate. The limitations are equally concrete. The test was opened before the propagation nuisance ranges were formally signed off. The positive class is simulated, so absolute recall depends on the impostor model and on how faithfully templates represent bursts; the real-burst proxy and the transfer checks bound, but do not remove, this dependence. Twenty sources per test set and 16 per fold leave wide intervals. The second machine reproduced the data bit for bit but placed model 6 about 0.05 recall below the original seeds, for reasons not yet isolated. Lenses are singular isothermal spheres.

X. Conclusion

A small Siamese network that aligns before it compares identifies lensed copies of real FRB spectra about twice as often as physics and feature baselines at a 1% false-positive rate, in a pre-registered blind test over all sources; it passes real-data checks designed for a detector without real positives, and it reproduces on a second machine. The same analysis shows that transit geometry, not classification accuracy, is the bottleneck for finding lensed repeaters, which points such detectors to wide-field radio arrays. All inputs are public (CHIME/FRB Catalog 2, DOI 10.11570/25.0066); code, frozen-set manifests, hashes and pre-registered plans are in the project repository (link withheld for review).

References

[1] D. R. Lorimer, M. Bailes, M. A. McLaughlin, D. J. Narkevic, and F. Crawford, "A bright millisecond radio burst of extragalactic origin," *Science*, vol. 318, p. 777, 2007.

[2] E. Petroff, J. W. T. Hessels, and D. R. Lorimer, "Fast radio bursts at the dawn of the 2020s," *A&A Rev.*, vol. 30, p. 2, 2022.

[3] Z.-X. Li, H. Gao, X.-H. Ding, G.-J. Wang, and B. Zhang, "Strongly lensed repeating fast radio bursts as precision probes of the universe," *Nature Communications*, vol. 9, p. 3833, 2018.

[4] O. Wucknitz, L. G. Spitler, and U.-L. Pen, "Cosmology with gravitationally lensed repeating fast radio bursts," *A&A*, vol. 645, p. A44, 2021.

[5] J. B. Muñoz, E. D. Kovetz, L. Dai, and M. Kamionkowski, "Lensing of fast radio bursts as a probe of compact dark matter," *Phys. Rev. Lett.*, vol. 117, p. 091301, 2016.

[6] C. Chang, S. Zhang *et al.*, "A lensed FRB candidate in the first CHIME/FRB catalogue and its potential implications," *arXiv:2406.19654*, 2024.

[7] Z. Kader *et al.*, "High-time resolution search for compact objects using fast radio burst gravitational lens interferometry with CHIME/FRB," *PhRvD*, vol. 106, p. 043016, 2022.

[8] C. Leung *et al.*, "Constraining primordial black holes using fast radio burst gravitational-lens interferometry with CHIME/FRB," *PhRvD*, vol. 106, p. 043017, 2022.

[9] K. Krochek and E. D. Kovetz, "Constraining primordial black hole dark matter with CHIME fast radio bursts," *PhRvD*, vol. 105, p. 103528, 2022.

[10] J. Bromley, I. Guyon, Y. LeCun, E. Säckinger, and R. Shah, "Signature verification using a "Siamese" time delay neural network," in *Advances in Neural Information Processing Systems 6*, 1993, pp. 737–744.

[11] R. Hadsell, S. Chopra, and Y. LeCun, "Dimensionality reduction by learning an invariant mapping," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2006, pp. 1735–1742.

[12] G. Koch, R. Zemel, and R. Salakhutdinov, "Siamese neural networks for one-shot image recognition," in *ICML Deep Learning Workshop*, 2015.

[13] S. Zagoruyko and N. Komodakis, "Learning to compare image patches via convolutional neural networks," in *Proc. IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 4353–4361.

[14] A. Dosovitskiy, P. Fischer, E. Ilg, P. Häusser, C. Hazirbas, V. Golkov, P. van der Smagt, D. Cremers, and T. Brox, "FlowNet: Learning optical flow with convolutional networks," in *Proc. IEEE International Conference on Computer Vision (ICCV)*, 2015, pp. 2758–2766.

[15] L. Bertinetto, J. Valmadre, J. F. Henriques, A. Vedaldi, and P. H. S. Torr, "Fully-convolutional Siamese networks for object tracking," in *Computer Vision – ECCV 2016 Workshops, LNCS 9914*, 2016, pp. 850–865.

[16] S. Goyal, D. Harikrishnan, S. J. Kapadia, and P. Ajith, "Rapid identification of strongly lensed gravitational-wave events with machine learning," *PhRvD*, vol. 104, p. 124057, 2021.

[17] F. Zhang, Q. Zhang, Q. Yang, J. Huang, Y. Yuan, and X. Fan, "Time-domain deep learning for pairwise identification of strongly lensed gravitational-wave candidates," *arXiv:2605.29510*, 2026.

[18] B. H. Chen, T. Hashimoto, T. Goto *et al.*, "Uncloaking hidden repeating fast radio bursts with unsupervised machine learning," *MNRAS*, vol. 509, p. 1227, 2022.

[19] J. Tobin, R. Fong, A. Ray, J. Schneider, W. Zaremba, and P. Abbeel, "Domain randomization for transferring deep neural networks from simulation to the real world," in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2017, pp. 23–30.

[20] CHIME/FRB Collaboration, "The second CHIME/FRB catalog of fast radio bursts," *ApJS, accepted; arXiv:2601.09399*, 2026, data release DOI 10.11570/25.0066.

[21] ——, "The CHIME fast radio burst project: System overview," *ApJ*, vol. 863, p. 48, 2018.

[22] L. Connor and V. Ravi, "Stellar prospects for FRB gravitational lensing," *MNRAS*, vol. 521, p. 4024, 2023.

[23] K. Vanderlinde *et al.*, "The canadian hydrogen observatory and radio-transient detector (CHORD)," *Canadian Long Range Plan for Astronomy and Astrophysics White Papers; arXiv:1911.01777*, 2019.

[24] S. C.-C. Ho *et al.*, "Future constraints on dark matter with gravitationally lensed fast radio bursts detected by BURSTT," *ApJ*, 2023.